

BC-760[B] / BC-760[R] / BC-780[R] Auto Hematology Analyzer
PPM (Planned Preventive Maintenance) Checklist

Hospital / Site Name		Department	
Analyzer Model		Serial Number	
Person in Charge		Telephone	
Service Date		Software Version	
Maintenance Interval	☐ 3 years / 72,000 tests ☐ 6 years / 144,000 tests		
Technician Name		Signature	

A. Maintenance Checking List

No.	Maintenance Item	Acceptance Criteria	Result
1	Check the waste cistern floater	No dirt on the WC2 floater	☐ Pass ☐ Fail ☐ N/A
2	Check the tube clamp	Clamp is free of deformation and faces the hole center of the test tube rack	☐ Pass ☐ Fail ☐ N/A
3	Check the tube rotary scanning mechanism	Slide rail is smooth without jamming; the test tube rotates steadily	☐ Pass ☐ Fail ☐ N/A
4	Check the tube detection sensor status	$0.6 \leq \text{change rate} \leq 1.2$	☐ Pass ☐ Fail ☐ N/A
5	Verify the sample probe position in the reaction bath	Sample probe is centered in the reaction bath	☐ Pass ☐ Fail ☐ N/A
6	Check the tube of the reagent cap assembly	Tube is normal — no bending or air leakage	☐ Pass ☐ Fail ☐ N/A
7	Clean the crystal on the solenoid valve	No crystal on the solenoid valve	☐ Pass ☐ Fail ☐ N/A
8	Clean the crystal on the syringe	No crystal on the syringe	☐ Pass ☐ Fail ☐ N/A
9	Clean the piercing probe, probe wipe and probe wipe barrier	Background results qualified after cleaning	☐ Pass ☐ Fail ☐ N/A
10	Check piercing probe abrasion status	Wear condition within acceptable limits	☐ Pass ☐ Fail ☐ N/A
11	Clean the covers of the HGB bath, RBC bath and optical reaction bath	No crystal or dust on the bath covers	☐ Pass ☐ Fail ☐ N/A
12	Lubricate the moving parts	Moving parts run smoothly without jamming	☐ Pass ☐ Fail ☐ N/A
13	Check and replace the wearing parts	Wearing parts replaced as required (see Parts table below)	☐ Pass ☐ Fail ☐ N/A

Wearing parts check / replacement (Item 13 detail):

Part	FRU Code	Replacement Cycle	Action
Piercing Needle (PN)	115-069029-00	120,000 times of piercing	☐ Checked ☐ Replaced ☐ N/A
Piercing Probe Wipe (PWD)	041-023727-00	120,000 times of piercing	☐ Checked ☐ Replaced ☐ N/A
Probe Wipe Filter (LF6)	115-011660-00	60,000 times / 3 years	☐ Checked ☐ Replaced ☐ N/A

B. Analyzer Status Checking List

Category	Item	Reference	Result
Pressure	PC (50 kPa)	45 ~ 60 kPa	☐ Pass ☐ Fail ☐ N/A
	SCI (40 kPa)	35 ~ 45 kPa	☐ Pass ☐ Fail ☐ N/A
	CBC_W (-30 kPa)	-90 ~ 0 kPa	☐ Pass ☐ Fail ☐ N/A
	WC2 (-40 kPa)	-45 ~ -35 kPa	☐ Pass ☐ Fail ☐ N/A
	Fluid pressure	50 ~ 120 kPa	☐ Pass ☐ Fail ☐ N/A
Temperature	SPMT temperature	31 ~ 38 °C	☐ Pass ☐ Fail ☐ N/A
	Ambient temperature	10 ~ 35 °C	☐ Pass ☐ Fail ☐ N/A
	DIFF / RET reaction bath	40 ~ 43 °C	☐ Pass ☐ Fail ☐ N/A
	Preheating bath	5 ~ 43 °C	☐ Pass ☐ Fail ☐ N/A
	RBC diluent	5 ~ 43 °C	☐ Pass ☐ Fail ☐ N/A
	Instrument temperature	10 ~ 41 °C	☐ Pass ☐ Fail ☐ N/A
	CPU temperature	0 ~ 100 °C	☐ Pass ☐ Fail ☐ N/A
	ESR assembly temperature	34 ~ 43 °C	☐ Pass ☐ Fail ☐ N/A
HGB blank voltage	HGB background voltage	3.20 ~ 4.80 V (adjustable)	☐ Pass ☐ Fail ☐ N/A
Sensor level settings	Fluorescent reagent detect sensor	2.10 ~ 2.60 V (adjustable)	☐ Pass ☐ Fail ☐ N/A
Cabling	Power cord, network cable, reagent tubes	Not bent or damaged, free of bubbles	☐ Pass ☐ Fail ☐ N/A
Waste tube	Waste tube	No crystal and no blockage	☐ Pass ☐ Fail ☐ N/A
Reagent container cap assembly	Air leakage; diluent sensor floater movement; sensor connectors	No air leakage; floater moves smoothly; connectors not loose	☐ Pass ☐ Fail ☐ N/A
Barcode scanner	Scanning function	Scanning is normal and operation is standard	☐ Pass ☐ Fail ☐ N/A

C. Post-Maintenance Performance Verification

Category	Parameter	Acceptance Reference	Result
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Background / blank — blood	WBC	$\leq 0.1 \times 10^9 / \text{L}$	☐ Pass ☐ Fail ☐ N/A
	WBC-D	$\leq 0.2 \times 10^9 / \text{L}$	☐ Pass ☐ Fail ☐ N/A
	RBC	$\leq 0.02 \times 10^{12} / \text{L}$	☐ Pass ☐ Fail ☐ N/A
	RBC-O	$\leq 0.02 \times 10^{12} / \text{L}$	☐ Pass ☐ Fail ☐ N/A
	HGB	$\leq 1 \text{ g} / \text{L}$	☐ Pass ☐ Fail ☐ N/A
	PLT	$\leq 5 \times 10^9 / \text{L}$	☐ Pass ☐ Fail ☐ N/A
	PLT-O	$\leq 5 \times 10^9 / \text{L}$	☐ Pass ☐ Fail ☐ N/A
Background / blank — body fluid	WBC-BF / TC-BF	$\leq 0.001 \times 10^9 / \text{L}$	☐ Pass ☐ Fail ☐ N/A
	RBC-BF	$\leq 0.003 \times 10^{12} / \text{L}$	☐ Pass ☐ Fail ☐ N/A
Scattergram	DIFF Channel	Graph normal and in proper position	☐ Pass ☐ Fail ☐ N/A
	RET Channel	Graph normal and in proper position	☐ Pass ☐ Fail ☐ N/A
Repeatability — whole blood	WBC / RBC / HGB / MCV / HCT / PLT	$\leq 2.5\% / \leq 1.5\% / \leq 1.0\% / \leq 1.0\% / \leq 1.5\% / \leq 4.0\%$	☐ Pass ☐ Fail ☐ N/A
Repeatability — body fluid	WBC-BF-TC-BF / RBC-BF	$\leq 30\% / \leq 40\%$ (or range $\leq 7000/\mu\text{L}$)	☐ Pass ☐ Fail ☐ N/A

Sign-off

Service Engineer / Technician Name: _____ Signature: _____ Date: _____	Customer / Site Representative Name: _____ Signature: _____ Date: _____
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Notes:

- Wear gloves during cleaning; wash hands with cleanser or disinfectant afterward. Probe Cleanser is a strong corrosive detergent — avoid skin/clothing contact.
- Only Mindray-supplied parts may be used for replacement.